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Supplementary Materials for

Discovery of novel antimicrobial peptides with notable antibacterial potency by an LLM-based foundation model

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Supplementary Information

Diversity and novelty analysis between Prompt, FT and CT

In order to conduct a comprehensive comparison among the prompt tuning, fine-tuning, and condition token models, we presented the match scores using statistical tables and cumulative distribution curves, as depicted in **Table S2** and **Fig. S1a-b**, respectively. **Table 2** provides a comprehensive statistical analysis of the mean match scores across different intervals. Notably, in the comparison of match scores with real AMPs, the average match scores for the Prompt, FT, and CT models were determined at 6.53, 15.58 and 13.12, respectively. Similarly, in the comparison of Internal Match Scores, the average match scores for the Prompt, FT, and CT models were observed to be 5.64, 13.80 and 11.63, respectively. These results offer a clearer understanding of the models' performance and highlight the varying degrees of similarity between the generated sequences and both Real-AMPs and internal sequences. Moreover, when examining **Fig. S1a-b**, it becomes apparent that across almost all intervals, the Prompt model consistently outperforms the FT and CT models in terms of match scores.

To further validate the diversity of AMPs generated by the three models, we computed the distribution of entropy, which reflects the uncertainty of the model's predictions at each time step. Higher entropy values indicate greater uncertainty and a larger number of possible outputs, while lower entropy values indicate higher certainty and fewer possible outputs for the model's predictions. As illustrated in **Fig. S2**, it is evident that the prompt model exhibits higher entropy compared to the FT and CT models. This indicates that the FT and CT models generate sequences with higher certainty compared to the prompt model, resulting in lower diversity of generated sequences.

The aforementioned findings provide compelling evidence of the significant distribution adjustment achieved through prompt tuning the original model. This adjustment ensures that the model, following prompt tuning, is capable of generating candidate sequences with desired activity. Moreover, compared to the fine-tuning and

condition token models, the prompt tuning approach generates AMPs with lower similarity to real AMPs, thereby exhibiting higher diversity. These results underscore the efficacy of prompt tuning in shaping the generated sequences, enhancing their dissimilarity to real AMPs, and promoting increased diversity in the generated AMPs.

Time and memory space usage

In this section, we present a comparative analysis of the memory space and time utilization of AMP-GPT AMP-prompt and AMP-Distillation. As shown in **Table S3**, the "Model Size" column represents the memory space usage occupied by each model, while the "Time" column indicates the time required to generate 5000 AMPs. The slower generation speed of the AMP-prompt model can be attributed to the concatenation of prompt embeddings at the beginning. In comparison to the other two models, the AMP-Distillation model has significantly smaller model size and nearly a tenfold improvement in speed compared to the AMP-Prompt model. The tests were conducted on a computer equipped with an AMD Ryzen 7 5800H CPU, an NVIDIA GeForce RTX 3060 Laptop GPU, and 16 GB of RAM.

Training information

The model is constructed using the huggingface's transformers package, and the hyperparameters of the model are as follows: "activation_function": "gelu_new", "architectures": ["GPT2LMHeadModel"], "attn_pdrop": 0.1, "n_ctx": 50, "n_embd": 768, "n_head": 8, "n_layer": 12. AdamW optimizer was used in training, for further details regarding additional parameters, please refer to our GitHub repository.

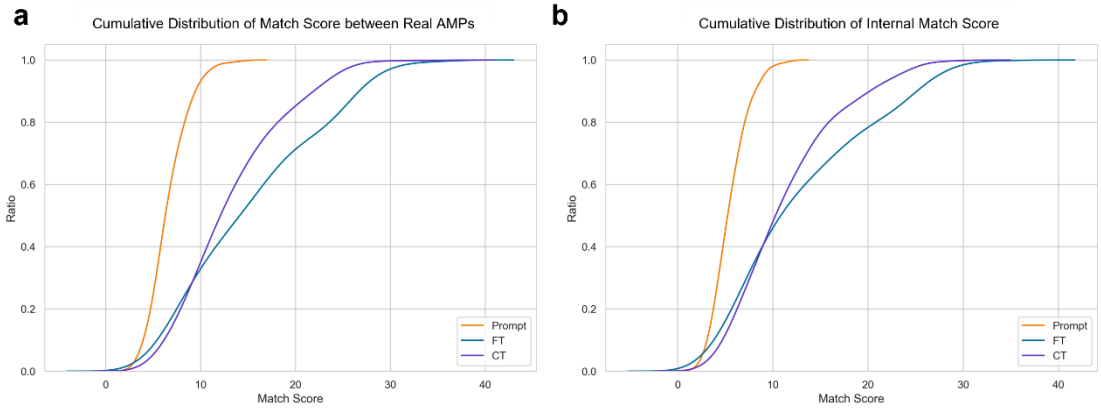


Fig. S1. Match scores cumulative distribution curves. Cumulative distribution curves of match score. **(a).** real AMPs and **(b).** them self.

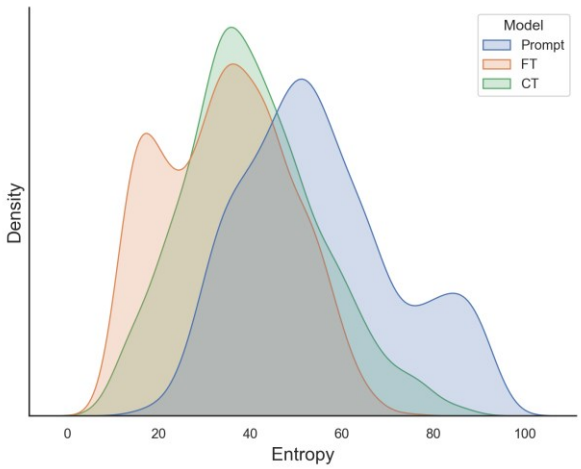


Fig. S2. The entropy distribution of Prompt, FT and CT.

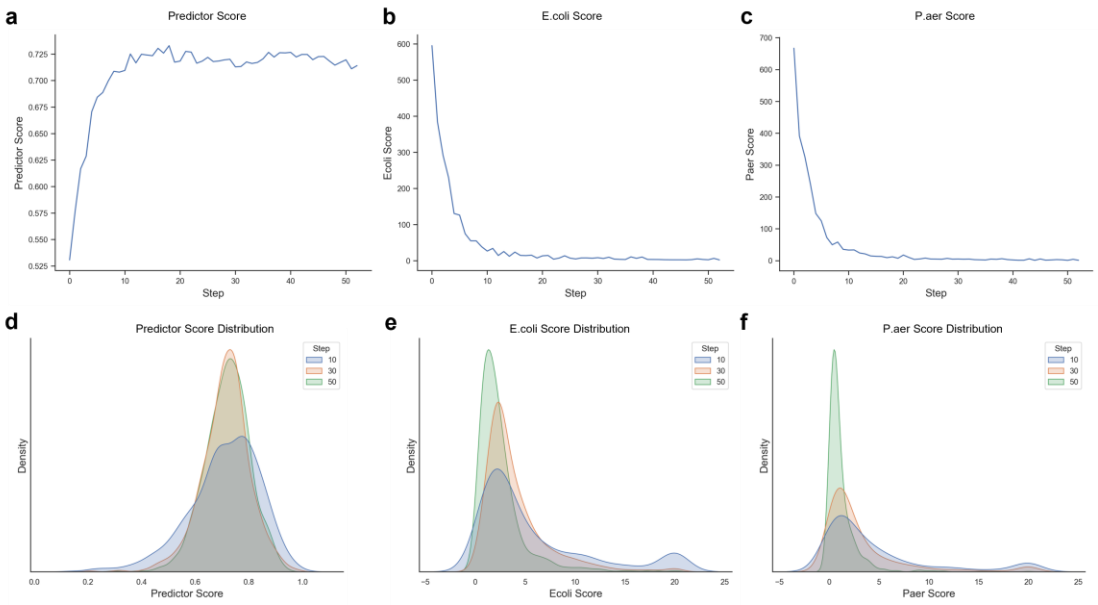


Fig. S3. The variations of different attributes during the RL. (a-c). The relationship between average score and step: **(a)** AMP predictor score, **(b)** *E. coli* predicted score and **(c)** *P. aeruginosa* predicted score. **(d)** AMP predictor score distribution. **(e)** *E. coli* predicted score distribution, and **(f)** *P. aeruginosa* predicted score distribution.

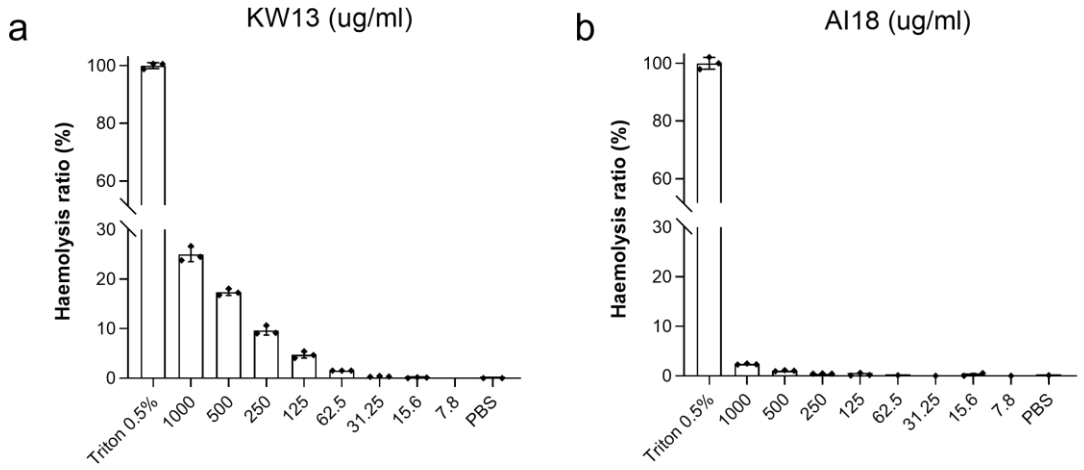


Fig. S4. Hemolytic concentration. 50% hemolytic concentration (HC50, 50% hemolysis) of KW13 and AI18.

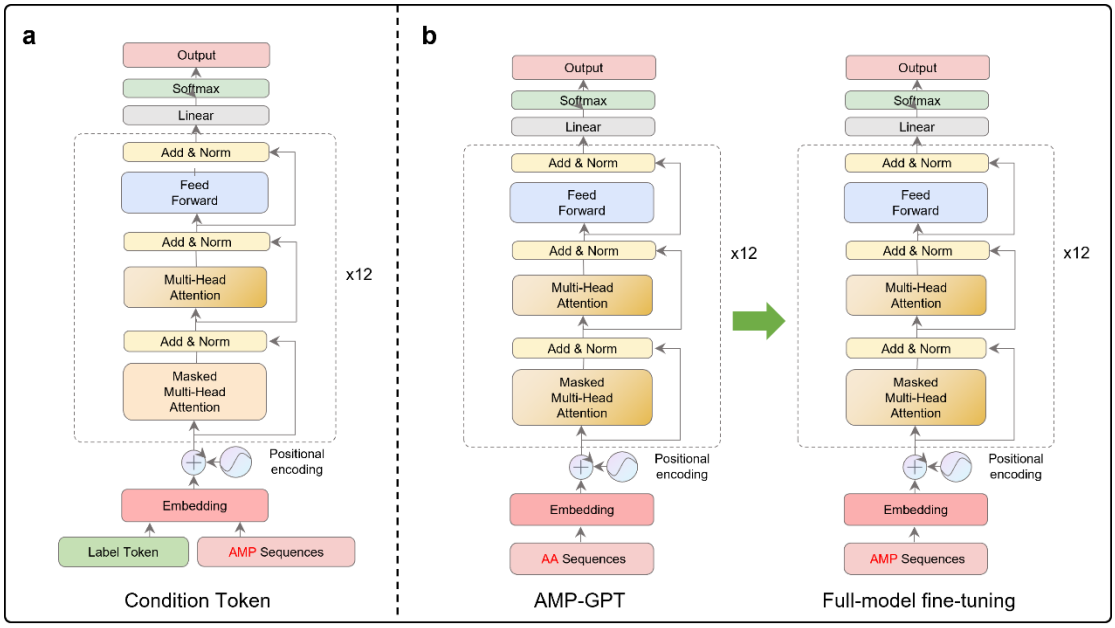


Fig. S5. The model architecture. (a). CT and **(b).** FT.

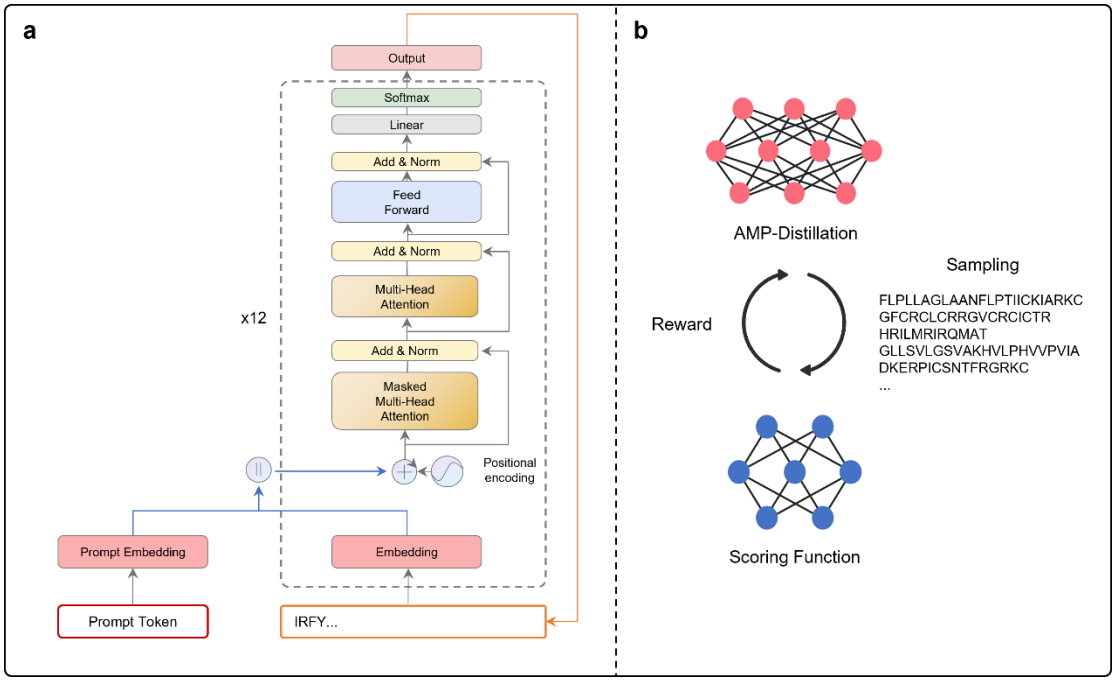


Fig. S6. The Inference stage. (a). AMP-Prompt and **(b).** RL.

Table S1. Percentage of the peptides identified as AMPs by all three predictors simultaneously.

Type	Percentage
UniProt	4.58%
GPT	5.13%
Prompt	61.8%
Prompt-TopK	83.4%
Distillation	59.2%
Real-AMPs	81.4%

Table S2. Match scores distribution.

	Model	Mean	Std	25%	50%	75%
Self	AMP-GPT	1.80	2.63	0.81	1.20	1.99
	AMP-Prompt	5.64	1.93	4.21	5.50	6.83
	AMP-Prompt-TopK	7.29	1.93	6.00	7.09	8.40
	AMP-Distillation	6.09	1.95	4.73	6.11	7.30
	Condition Token (CT)	11.63	5.68	7.21	10.47	14.75
	Fine-Tuning (FT)	13.80	7.96	7.09	11.71	19.67
Real AMP	AMP-GPT	4.17	3.19	2.79	3.53	4.43
	AMP-Prompt	6.53	2.13	5.08	6.23	7.87
	AMP-Prompt-TopK	7.43	2.15	5.89	7.11	8.62
	AMP-Distillation	6.89	2.13	5.38	6.60	8.11
	Condition Token (CT)	13.12	5.72	8.65	12.14	16.61
	Fine-Tuning (FT)	15.58	7.90	8.40	14.59	23.05

Table S3. Time and space comparison of AMP-Distillation and AMP-prompt and AMP-GPT using.

Methods	Model Size(MB)	Time(S)
AMP-GPT	334	51.9
AMP-prompt	335	136.8
AMP-Distillation	16	11.9

Table S4. Reward of RL.

Reward	Detail
Classification probably (CP)	Classification probably predicted by Macrel and divided into 5 bins.
MIC of three bacteria (MIC)	MIC predicted by AMP-MIC divided into 10 bins.
Sequence Length (SL)	Sequence length of peptides.
Memory Similarity (MS)	Similarity between generated sequences and AMPs in memory queue.
Charge	Continuous value.

Table S5. The performance of the MIC predictors.

Bacteria	Metric	RNN	GPT	AMP-MIC
S. aureus	MSE	0.424	0.394	**0.373**
	R^2	0.792	0.801	**0.811**
E. coli	MSE	0.464	0.413	0.393
	R^2	0.771	0.796	**0.807**
P. aeruginosa	MSE	0.323	0.328	**0.263**
	R^2	0.816	0.813	**0.851**

Table S6. Antimicrobial susceptibility testing results of clinical bacteria.

Antibiotic	MIC（μg/mL）and susceptibility profile					
	316039	304238	103231	327004	325016	418015
Ticarcillin/clavulanic acid	128 R	128 R	32 R	128 R	128 R	128 R
Piperacillin/tazobactam	128 R	ND	128 R	128 R	128 R	128 R
Ceftazidime	64 R	32 R	32 R	64 R	64 R	64 R
Ceftriaxone	ND	ND	64 R	ND	ND	ND
Cefoperazone/sulbactam	64 R	64 R	64 R	64 R	64 R	64 R
Cefepime	32 R	16 I	32 R	32 R	32 R	32 R
Aztreonam	ND	ND	ND	64 R	64 R	64 R
Imipenem	16 R	8 R	0.25 S	16 R	16 R	16 R
Meropenem	16 R	4 I	ND	16 R	16 R	16 R
Amikacin	ND	2 S	16 S	64 R	64 R	64 R
Tobramycin	1 S	1 S	ND	16 R	16 R	16 R
Ciprofloxacin	4 R	1 I	ND	4 R	4 R	4 R
Levofloxacin	4 I	4 R	8 R	8 R	8 R	8 R
Doxycycline	16 R	ND	ND	2 S	16 R	8 I
Minocycline	4 S	ND	ND	4 S	16 R	16 R
Colistin	0.5 S	0.5 S	0.5 S	0.5 S	0.5 S	16 R
Sulfamethoxazole/trimethoprim	160 R	320 R	320 R	320 R	20 S	320 R
Tigecycline	2 S	-	0.5 S	0.5 S	8 R	2 R

R, resistant; S, sensitive; I, intermediate. -, intrinsic resistance. ND, not determined.

Table S7. The comparison with AMPs in analogous work.

Method	ID	MIC[a] (μg/ml)			Coverage%	Similarity%	E-value	BLAST score
		S. aureus	*E. coli*	*P. aeruginosa*				
Our	RR13	64	16	32	53.8	100	0.35	30.33
	AI18	32	8	8	77.8	71.4	1.07	31.60
	KW20	16	4	8	65.0	76.9	0.61	30.75
	KW13	8	4	4	84.6	75.0	0.30	30.33
	VK18	16	16	16	55.6	80.0	0.15	32.03
	SK18	8	8	16	83.3	80.0	0.47	30.75
	RV15	8	8	16	73.3	72.7	1.05	29.91
	IR20	32	32	64	80.0	63.2	0.07	33.72
	RK17	32	32	16	64.7	81.8	0.80	32.03
	LW14	16	64	16	78.6	69.2	0.01	34.57
Das et al. [30b]	YI12	8	31	125	66.7	87.5	1.20	29.90
	FK13	16	31	63	69.2	88.9	0.03	32.88
Huang et al.[63c]	CRRI3	8	64	16	100	100	0.87	30.33
	CRRI4	8	64	64	100	100	0.12	30.33
	CRRI7	8	32	32	100	100	0.03	32.03

[a] Minimal inhibitory concentration; [b] *Staphylococcus aureus* (ATCC 29737), *Escherichia coli* (ATCC 25922), *Pseudomonas aeruginosa* (ATCC 9027); [c] *Staphylococcus aureus* (ATCC 6538), *Escherichia coli* (ATCC 1806), *Pseudomonas aeruginosa* (ATCC 2512).

Table S8. The abbreviations of AMPs and their corresponding sequences.

Abbreviation	Sequence	Expected MS	Measured MS
RR13	RRWFKIRMAAKLA-NH2	1646.08	1645.88
AI18	AIPKRLRRFYLRALARRL-NH2	2268.84	2269.44
KW20	KWRKVRAKFWRKWLAGLINT-NH2	2557.14	2557.84
KW13	KWKKWVRAIKKLV-NH2	1682.18	1682.40
VK18	VKRFRRWWKPWRKILHLV-NH2	2505.13	2505.32
SK18	SKLWKKIVKALKAALKTL-NH2	2038.64	2039.00
RV15	RVLWIKRWIKRFFRP-NH2	2100.64	2101.00
IR20	IRFYWRRVQVWRGIWRRLVR-NH2	2801.40	2801.20
RK17	RKKWWAYLLAKIAKKVK-NH2	2129.72	2129.94
LW14	LWRFKRWWWWKKIL-NH2	2132.65	2132.80
GI16	GIKKFAKLLKYIAGKL-NH2	1790.29	1790.55
RR16	RRWGKLIKKIAKKFGG-NH2	1885.35	1885.00
YK19	YKWVANVAKKLIKLLKVLK-NH2	2254.89	2254.80
AK15	AKKFKIWGAIKRLLA-NH2	1742.20	1741.95
KK12	KKFLKLIKGIKR-NH2	1470.93	1471.05
KWI18	KWKWPPRWPPWRPVWKVI-NH2	2441.96	2441.85
LF19	LFKVIFGIVKKKKLLPKFF-NH2	2292.98	2293.00
AV20	AVWRWLWKGSLAKGIIKFLK-NH2	2399.96	2400.40
FI19	FIKKLLKLLGKLARLAHAL-NH2	2145.80	2146.16
IF20	IFKLLKKIIKKIVFKIAKAL-NH2	2355.18	2355.16
RW20	RWPFKWKRPRILKLICFRAL-NH2	2628.34	2628.80
RW12	RWKVIRRRWLRL-NH2	1738.18	1738.20
KW18	KWKWWKRRKFCLLYIVFC-NH2	2504.16	2504.40